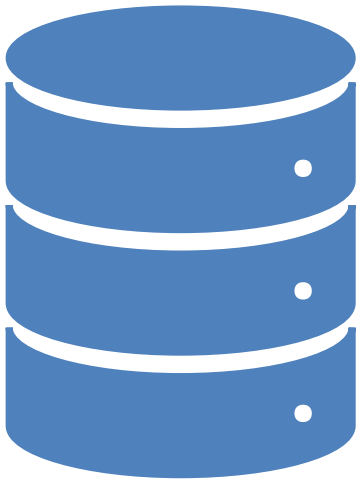


INTRODUCTION TO SQL

What is SQL?

- **Structured Query Language**
- SQL is Structured Query Language, which is a computer language for **storing**, **manipulating** and **retrieving** data stored in relational database.

- SQL is the standard language for **Relation** Database System. All relational database management systems like "MySQL, MS Access, Oracle, Sybase, Informix, postgres and SQL Server" use SQL as standard database language.

SQL Commands

- **DDL - Data Definition Language**
- **DML - Data Manipulation Language**
- **DCL - Data Control Language**
- **DQL - Data Query Language**

SQL Commands

DDL - Data Definition Language:

Command	Description
CREATE	Creates a new table, a view of a table, or other object in database
ALTER	Modifies an existing database object, such as a table.
DROP	Deletes an entire table, a view of a table or other object in the database.

DML - Data Manipulation Language:

Command	Description
INSERT	Creates a record
UPDATE	Modifies records
DELETE	Deletes records

SQL Commands

DCL - Data Control Language:

Command	Description
GRANT	Gives a privilege to user
REVOKE	Takes back privileges granted from user

DQL - Data Query Language:

Command	Description
SELECT	Retrieves certain records from one or more tables

SQL RDBMS Concepts

- TABLE
  - RECORD
  - COLUMN
  - CELL
- NULL?

ID	NAME	AGE	ADDRESS	SALARY
1	Ramesh	32	Ahmedabad	2000.00
2	Khilan	25	Delhi	1500.00
3	kaushik	23	Kota	2000.00
4	Chaitali	25	Mumbai	6500.00
5	Hardik	27	Bhopal	8500.00
6	Komal	22	MP	4500.00
7	Muffy	24	Indore	10000.00

ADDRESS
Ahmedabad
Delhi
Kota
Mumbai
Bhopal
MP
Indore

1	Ramesh	32	Ahmedabad	2000.00

SQL RDBMS Concepts

- SQL Constraints: (applied on columns)
  - NOT NULL Constraint
  - UNIQUE Constraint
  - PRIMARY Key
  - FOREIGN Key
  - CHECK Constraint
- Data Integrity:
  - Entity Integrity: There are no duplicate rows in a table
  - Domain Integrity: Enforces valid entries for a given column by restricting the type
  - Referential Integrity: Rows cannot be deleted which are used by other records
  - User-Defined Integrity: Enforces some specific business rules

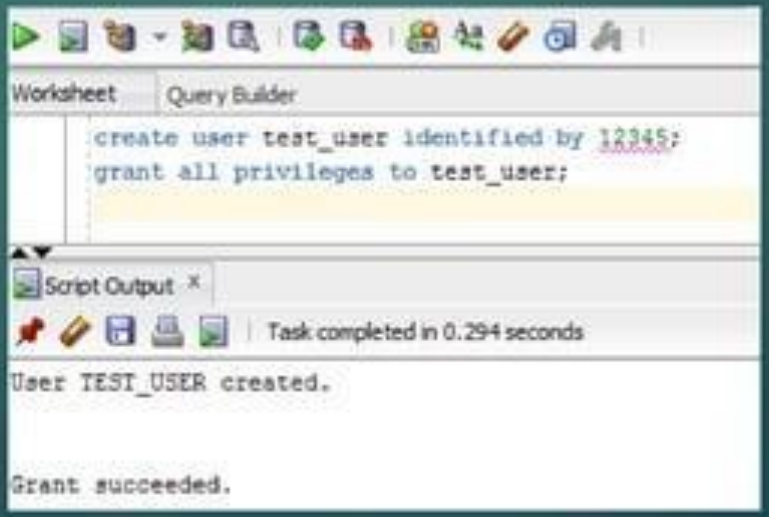
SQL RDBMS Concepts

- Data Types:
- Character datatypes:
  - CHAR
  - NCHAR
  - NVARCHAR2
  - VARCHAR2
- *Numeric* datatypes:
  - NUMBER
  - BINARY_FLOAT
  - BINARY_DOUBLE
- *Date time* datatype:
  - DATE

DCL: CREATE USER

- 1) Define User with Username/Password
- 2) Grants Sufficient Privileges.

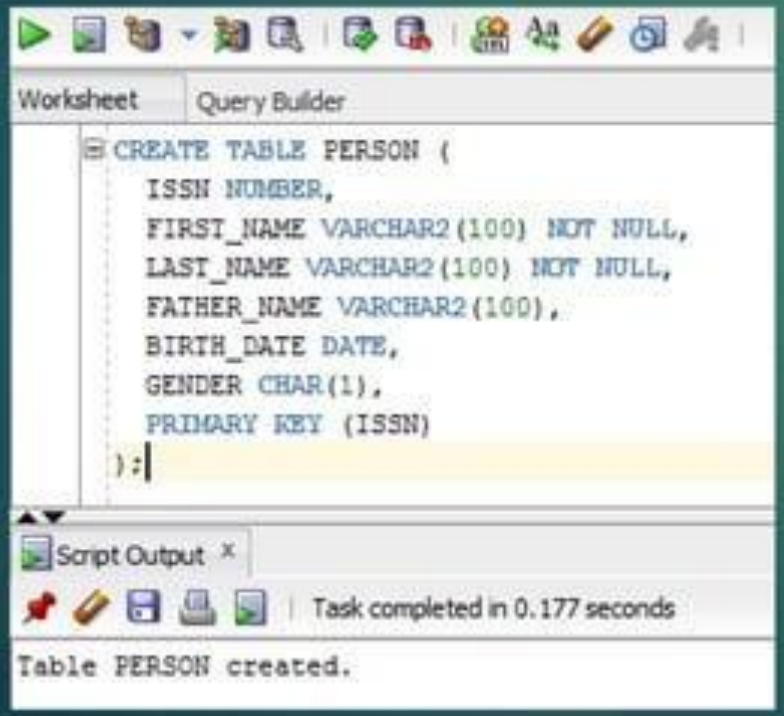
- Using SQL Developer (UI, Command)
- Using SQL *PLUS (Command)

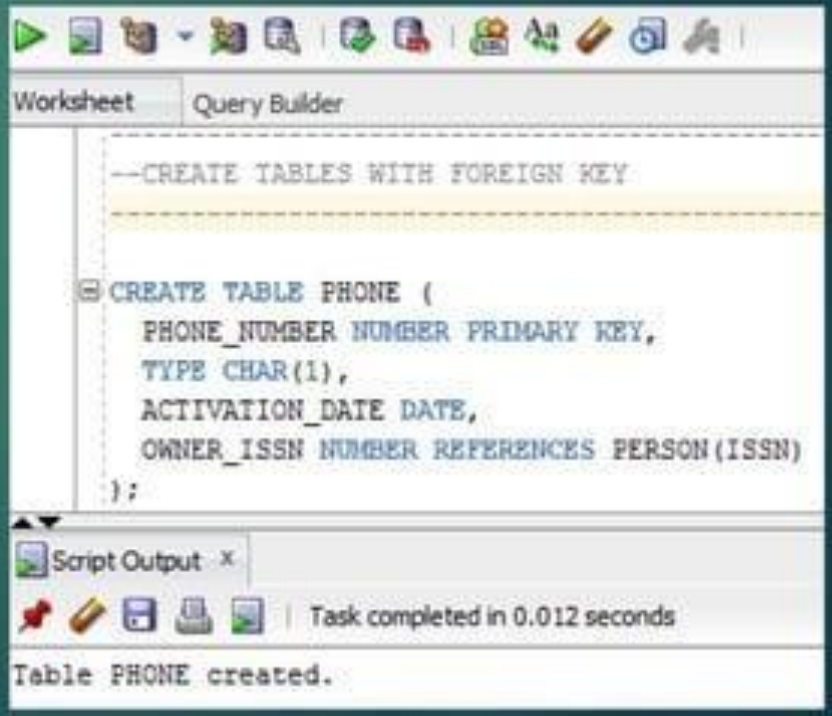


SQL : DDL

- Create Table:
  - COLUMNS
  - CONSTRAINTS

```
CREATE TABLE table_name(
   column1 datatype,
   column2 datatype,
   column3 datatype,
   .....
   columnN datatype,
   PRIMARY KEY( one or more columns )
);
```





SQL: DDL

- DROP TABLE:

```
--DROP TABLES
DROP TABLE PHONE;
```

Script Output

Task completed in 0.247 seconds

```
Table PHONE dropped.
```

- ALTER TABLE

```
--ALTER TABLE
ALTER TABLE PHONE
ADD STATUS CHAR(1);

ALTER TABLE PHONE
DROP COLUMN STATUS;
```

- OTHER DDL COMMANDS!!!!

SQL: DML: INSERT

- INSERT INTO:
  - NUMBER, CHAR/VARCHAR2, DATE?

```
INSERT INTO PERSON VALUES (1,'HOSSEIN','ASKARIAN','ALI',TO_DATE('12-JUN-1990'),'M');
INSERT INTO PERSON VALUES (2,'SADEGH','KARIMI','HOSSEIN',TO_DATE('05/04/1990','DD/MM/YYYY'),'M');
INSERT INTO PERSON VALUES (3,'HOSSEIN','ASKARIAN','ALI',TO_DATE('10-04-1367','DD-MM-YYYY','NLS_CALENDAR=PERSIAN'),'F');
```

Script Output | Task completed in 0.055 seconds

```
1 row inserted.

1 row inserted.
```

```
-- WRONG: PK?
INSERT INTO PERSON VALUES (1,'EHSAN','HAMZEI','ALIREZA',TO_DATE('26-JUN-1991'),'M');
```

Script Output | Query Result | Task completed in 0.096 seconds

```
Error starting at line : 51 in command -
INSERT INTO PERSON VALUES (1,'EHSAN','HAMZEI','ALIREZA',TO_DATE('26-JUN-1991'),'M')
Error report -
SQL Error: ORA-00001: unique constraint (TEST_USER.SYS_C0010353) violated
00001. 00000 -  "unique constraint (%s.%s) violated"
*Cause:    An UPDATE or INSERT statement attempted to insert a duplicate key.
           For Trusted Oracle configured in DBMS MAC mode, you may see
           this message if a duplicate entry exists at a different level.
*Action:   Either remove the unique restriction or do not insert the key.
```

SQL: DML: INSERT

- INSERT INTO:
  - SPECIFIC COLUMNS

```
-- SPECIFIC COLUMNS
INSERT INTO PERSON (BIRTH_DATE,FATHER_NAME,LAST_NAME,FIRST_NAME,ISSN)
VALUES (TO_DATE('12021990','DDMMYYYY'),'HOSSEIN','HEMMATI','ALIREZA',4);
```

```
-- WRONG: NOT NULL?
INSERT INTO PERSON (BIRTH_DATE,FATHER_NAME,FIRST_NAME,ISSN)
VALUES (TO_DATE('12021990','DDMMYYYY'),'SADEGH','SETAREH',5);
```

Script Output × | Query Result ×

Task completed in 0.011 seconds

```
Error starting at line : 57 in command -
INSERT INTO PERSON (BIRTH_DATE,FATHER_NAME,FIRST_NAME,ISSN)
VALUES (TO_DATE('12021990','DDMMYYYY'),'SADEGH','SETAREH',5)
Error report -
SQL Error: ORA-01400: cannot insert NULL into ("TEST_USER"."PERSON"."LAST_NAME")
01400. 00000 -  "cannot insert NULL into (%s)"
*Cause:    An attempt was made to insert NULL into previously listed objects.
*Action:   These objects cannot accept NULL values.
```

SQL: DML: INSERT

- INSERT INTO:
  - FOREIGN KEY

```
-- FOREIGN KEY
INSERT INTO PHONE VALUES (09324545321,'M',TO_DATE('01-01-1395','DD-MM-YYYY','NLS_CALENDAR = PERSIAN'),1);
-- WRONG: FK?
INSERT INTO PHONE VALUES (02134324533,'T',TO_DATE('03-05-1392','DD-MM-YYYY','NLS_CALENDAR = PERSIAN'),11)
```

Script Output | Query Result

Task completed in 0.03 seconds

```
1 row inserted.

Error starting at line : 64 in command -
INSERT INTO PHONE VALUES (02134324533,'T',TO_DATE('03-05-1392','DD-MM-YYYY','NLS_CALENDAR = PERSIAN'),11)
Error report -
SQL Error: ORA-02291: integrity constraint (TEST_USER.SYS_C0010357) violated - parent key not found
02291. 00000 - "integrity constraint (%s.%s) violated - parent key not found"
*Cause:    A foreign key value has no matching primary key value.
*Action:   Delete the foreign key or add a matching primary key.
```

- ALL DML COMMANDS NEED COMMIT

SQL: DML: UPDATE

- UPDATE

```
UPDATE PERSON
SET FATHER_NAME = 'AAAAAA';
-- ROLLBACK!
ROLLBACK;
```

Script Output × Query Result ×

Task completed in 0.003 seconds

```
4 rows updated.

Rollback complete.
```

- WHERE CLAUSE

```
--USING WHERE CLAUSE
UPDATE PERSON
SET FATHER_NAME = 'ALIREZA'
WHERE ISSN = 1;
COMMIT;
```

Script Output × Query Result ×

Task completed in 0.001 seconds

```
1 row updated.

Commit complete.
```

SQL: DML: DELETE

- DELETE (DELETE VS DROP????)

```
DELETE PERSON;
```

Script Output | Query Result

Task completed in 0.01 seconds

```
Error starting at line : 85 in command -
DELETE PERSON
Error report -
SQL Error: ORA-02292: integrity constraint (TEST_USER.SYS_C0010357) violated - child record found
02292. 00000 - "integrity constraint (%s.%s) violated - child record found"
*Cause:    attempted to delete a parent key value that had a foreign
           dependency.
*Action:   delete dependencies first then parent or disable constraint.
```

```
DELETE PERSON WHERE ISSN = 2;
ROLLBACK;
```

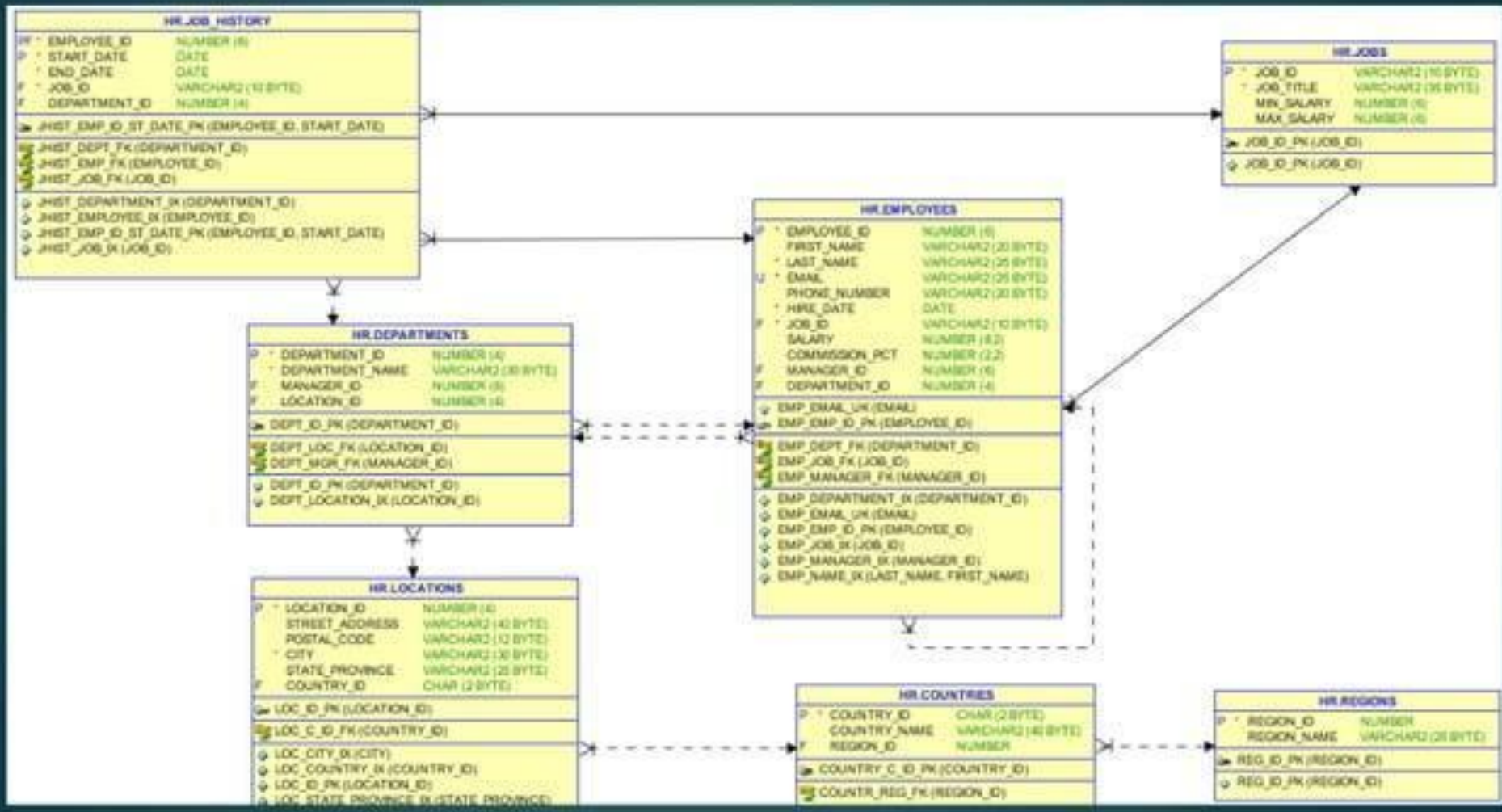
Script Output | Query Result

Task completed in 0.001 seconds

```
1 row deleted.

Rollback complete.
```

HR-Schema

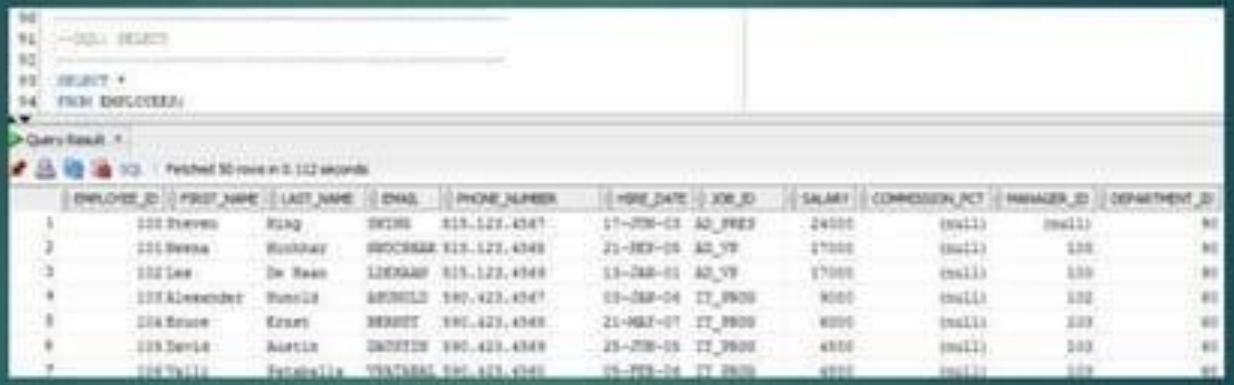

HR.JOB_HISTORY
EMPLOYEE_ID NUMBER (6)
START_DATE DATE
END_DATE DATE
JOB_ID VARCHAR2 (10 BYTE)
DEPARTMENT_ID NUMBER (4)
JHIST_EMP_ID_ST_DATE_PK (EMPLOYEE_ID, START_DATE)
JHIST_DEPT_FK (DEPARTMENT_ID)
JHIST_EMP_FK (EMPLOYEE_ID)
JHIST_JOB_FK (JOB_ID)
JHIST_DEPARTMENT_IX (DEPARTMENT_ID)
JHIST_EMPLOYEE_IX (EMPLOYEE_ID)
JHIST_EMP_ID_ST_DATE_PK (EMPLOYEE_ID, START_DATE)
JHIST_JOB_IX (JOB_ID)
HR.JOBS
JOB_ID VARCHAR2 (10 BYTE)
JOB_TITLE VARCHAR2 (35 BYTE)
MIN_SALARY NUMBER (6)
MAX_SALARY NUMBER (6)
JOB_ID_PK (JOB_ID)
JOB_ID_PK (JOB_ID)
HR.EMPLOYEES
EMPLOYEE_ID NUMBER (6)
FIRST_NAME VARCHAR2 (20 BYTE)
LAST_NAME VARCHAR2 (25 BYTE)
EMAIL VARCHAR2 (25 BYTE)
PHONE_NUMBER VARCHAR2 (20 BYTE)
HIRE_DATE DATE
JOB_ID VARCHAR2 (10 BYTE)
SALARY NUMBER (8,2)
COMMISSION_PCT NUMBER (2,2)
MANAGER_ID NUMBER (6)
DEPARTMENT_ID NUMBER (4)
EMP_EMAIL_UK (EMAIL)
EMP_EMP_ID_PK (EMPLOYEE_ID)
EMP_DEPT_FK (DEPARTMENT_ID)
EMP_JOB_FK (JOB_ID)
EMP_MANAGER_FK (MANAGER_ID)
EMP_DEPARTMENT_IX (DEPARTMENT_ID)
EMP_EMAIL_UK (EMAIL)
EMP_EMP_ID_PK (EMPLOYEE_ID)
EMP_JOB_IX (JOB_ID)
EMP_MANAGER_IX (MANAGER_ID)
EMP_NAME_IX (LAST_NAME, FIRST_NAME)
HR.DEPARTMENTS
DEPARTMENT_ID NUMBER (4)
DEPARTMENT_NAME VARCHAR2 (30 BYTE)
MANAGER_ID NUMBER (6)
LOCATION_ID NUMBER (4)
DEPT_ID_PK (DEPARTMENT_ID)
DEPT_LOC_FK (LOCATION_ID)
DEPT_MGR_FK (MANAGER_ID)
DEPT_ID_PK (DEPARTMENT_ID)
DEPT_LOCATION_IX (LOCATION_ID)
HR.LOCATIONS
LOCATION_ID NUMBER (4)
STREET_ADDRESS VARCHAR2 (40 BYTE)
POSTAL_CODE VARCHAR2 (12 BYTE)
CITY VARCHAR2 (30 BYTE)
STATE_PROVINCE VARCHAR2 (25 BYTE)
COUNTRY_ID CHAR (2 BYTE)
LOC_ID_PK (LOCATION_ID)
LOC_C_ID_FK (COUNTRY_ID)
LOC_CITY_IX (CITY)
LOC_COUNTRY_IX (COUNTRY_ID)
LOC_ID_PK (LOCATION_ID)
LOC_STATE_PROVINCE_IX (STATE_PROVINCE)
HR.COUNTRIES
COUNTRY_ID CHAR (2 BYTE)
COUNTRY_NAME VARCHAR2 (40 BYTE)
REGION_ID NUMBER
COUNTRY_C_ID_PK (COUNTRY_ID)
COUNTR_REG_FK (REGION_ID)
HR.REGIONS
REGION_ID NUMBER
REGION_NAME VARCHAR2 (25 BYTE)
REG_ID_PK (REGION_ID)
REG_ID_PK (REGION_ID)

SQL: DQL: SELECT

- SELECT

```
SELECT column1, column2....columnN
FROM   table_name;
```

  - *



  - SPECIFIC COLUMNS



```
96  SELECT FIRST_NAME, LAST_NAME
97  FROM EMPLOYEES;
```

Query Result | SQL | Fetched 50 rows in 0.0

	FIRST_NAME	LAST_NAME
1	Ellen	Abel
2	Sundar	Ande
3	Mozhe	Atkinson
4	David	Austin
5	Hermann	Baer
6	Shelli	Baida
7	Amit	Banda
8	Elizabeth	Bates
9	Sarah	Bell

SQL: DQL: SELECT

- DISTINCT CLUASE

```
SELECT DISTINCT column1, column2....columnN
FROM   table_name;
```

```
99  SELECT DEPARTMENT_ID
100 FROM EMPLOYEES
```

Query Result | SQL | Fetched 50

	DEPARTMENT_ID
1	90
2	90
3	90
4	60
5	60
6	60
7	60
8	60
9	100
10	100

```
99  SELECT DISTINCT DEPARTMENT_ID
100 FROM EMPLOYEES
```

Query Result | SQL | All Rows Fetched: 12 in

	DEPARTMENT_ID
1	100
2	30
3	(null)
4	90
5	20
6	70
7	110
8	50
9	80
10	40

SQL: DQL: SELECT

- WHERE CLUASE

```
SELECT column1, column2....columnN
FROM   table_name
WHERE  CONDITION;
```

```
SELECT *
FROM EMPLOYEES
WHERE DEPARTMENT_ID = 100;
```

Query Result

All Rows Fetched: 6 in 0.198 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
2	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
3	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
4	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100
5	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100
6	113	Luis	Popp	LPOPP	515.124.4567	07-DEC-07	FI_ACCOUNT	6900	(null)	108	100

```
SELECT column1, column2....columnN
FROM   table_name
WHERE  CONDITION-1 {AND|OR} CONDITION-2;
```

```
SELECT *
FROM EMPLOYEES
WHERE DEPARTMENT_ID = 100 AND FIRST_NAME LIKE 'J%';
```

Query Result

All Rows Fetched: 2 in 0.004 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
2	112	Jose Manuel	Urman	JMURMAN	515.124.4469	07-MAR-06	FI_ACCOUNT	7800	(null)	108	100

SQL: DQL: SELECT

- IN CLUASE
  - NOT IN

```
SELECT column1, column2....columnN
FROM   table_name
WHERE  column_name IN (val-1, val-2,...val-N);
```

```
110  SELECT *
111  FROM EMPLOYEES
112  WHERE DEPARTMENT_ID IN (100,90);
```

Query Result

SQL | All Rows Fetched: 9 in 0.003 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
4	108	Nancy	Greenberg	NGREENBE	515.124.4569	17-AUG-02	FI_MGR	12008	(null)	101	100
5	109	Daniel	Faviet	DFAVIET	515.124.4169	16-AUG-02	FI_ACCOUNT	9000	(null)	108	100
6	110	John	Chen	JCHEN	515.124.4269	28-SEP-05	FI_ACCOUNT	8200	(null)	108	100
7	111	Ismael	Sciarra	ISCIARRA	515.124.4369	30-SEP-05	FI_ACCOUNT	7700	(null)	108	100

- NULL IN WHERE?
  - IS NULL
  - IS NOT NULL

```
114  -- WRONG: USE IS NULL/ IS NOT NULL
115  SELECT *
116  FROM EMPLOYEES
117  WHERE DEPARTMENT_ID = NULL;
118
119  -- WRONG: USE IS NULL/ IS NOT NULL
120  SELECT *
121  FROM EMPLOYEES
122  WHERE DEPARTMENT_ID <> NULL;
123
124  SELECT *
125  FROM EMPLOYEES
126  WHERE DEPARTMENT_ID IS NULL;
127
128  SELECT *
129  FROM EMPLOYEES
130  WHERE DEPARTMENT_ID IS NOT NULL;
```

SQL: DQL: SELECT

- ORDER BY

```
SELECT column1, column2....columnN
FROM   table_name
WHERE  CONDITION
ORDER BY column_name {ASC|DESC};
```

```
143 SELECT *
144   FROM EMPLOYEES
145   WHERE DEPARTMENT_ID = 50
146   ORDER BY SALARY;
```

Explain Plan | Query Result

SQL | All Rows Fetched: 45 in 0.014 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	132	TJ	Olson	TJOLSON	650.124.8234	10-APR-07	ST_CLERK	2100	(null)	121	50
2	136	Hazel	Philtanker	HPHILTAN	650.127.1634	06-FEB-08	ST_CLERK	2200	(null)	122	50
3	128	Steven	Markle	SMARKLE	650.124.1434	08-MAR-08	ST_CLERK	2200	(null)	120	50
4	135	Ki	Gee	KGEE	650.127.1734	12-DEC-07	ST_CLERK	2400	(null)	122	50
5	127	James	Landry	JLANDRY	650.124.1334	14-JAN-07	ST_CLERK	2400	(null)	120	50
6	140	Joshua	Patel	JPATEL	650.121.1834	06-APR-06	ST_CLERK	2500	(null)	123	50

SQL: DQL: SELECT

- GROUP BY
- AGGREGATE FUCTION
  - MAX,MIN
  - AVG
  - COUNT
  - SUM
  - ...
- HAVING CLAUSE

```
177  SELECT DEPARTMENT_ID, COUNT(*) AS NUM_EMPS, AVG(SALARY) AS  AVG_SALARY
178  FROM EMPLOYEES
179  GROUP BY DEPARTMENT_ID;
```

Explain Plan | Query Result

SQL | All Rows Fetched: 12 in 0.007 seconds

	DEPARTMENT_ID	NUM_EMPS	AVG_SALARY
1	100	6	8601.333333333333333333333333333333333333
2	30	6	4150
3	(null)	1	7000
4	90	3	19333.333333333333333333333333333333333333
5	20	2	9500
6	70	1	10000
7	110	2	10154
8	50	45	3475.555555555555555555555555555555555556

SQL: DQL: SELECT

- SELECT FROM MULTIPLE TABLES
  - CARTESIAN MULTIPICATION

```
212  --CARTESIAN MULTIPICATIONS
213  SELECT E.*,J.JOB_ID AS N_JOB_ID
214  FROM EMPLOYEES E, JOBS J;
215
216  SELECT COUNT(*) AS TABLE_DIM, COUNT(DISTINCT E.EMPLOYEE_ID) AS EMP_DIM, COUNT(DISTINCT J.JOB_ID) AS JOB_DIM
217  SELECT COUNT(*) AS TABLE_DIM, COUNT(DISTINCT E.EMPLOYEE_ID) AS EMP_DIM, COUNT(DISTINCT J.JOB_ID) AS JOB_DIM
218  FROM EMPLOYEES E, JOBS J;
```

  - JOIN

```
--JOIN
SELECT COUNT(*)
FROM EMPLOYEES E, JOBS J
WHERE E.JOB_ID = J.JOB_ID;

SELECT COUNT(*)
FROM EMPLOYEES E
JOIN JOBS J ON E.JOB_ID = J.JOB_ID;
```



THANK YOU!